# Hypotonia, Ataxia, Developmental Delay, and Tooth Enamel Defect Syndrome (HADDTS)

### A CTBP1-related neurodevelopmental disorder — comprehensive disease characterization

## Summary (Answer to the Research Question)

**Hypotonia, Ataxia, Developmental Delay, and Tooth Enamel Defect Syndrome (HADDTS)** is an ultra-rare, autosomal dominant neurodevelopmental disorder caused by **de novo heterozygous pathogenic variants in *CTBP1*** (C-terminal binding protein 1; HGNC:2494), a NAD(H)-dependent transcriptional corepressor that also acts as a cytoplasmic membrane-fission factor (BARS). The overwhelming majority of patients (~90%) carry a single recurrent missense hotspot, **c.991C>T / c.1024C>T, p.Arg342Trp** (transcript-dependent numbering; originally reported as p.R331W), located in the C-terminal **PLDLS-binding cleft** through which CtBP1 recruits chromatin-modifying corepressor partners. The mutation acts as a **dominant-negative / interfering allele** (not simple haploinsufficiency), impairing assembly of CtBP–HDAC1/2–LSD1 repressor complexes that guide corticogenesis, deregulating transcription, sensitizing cells to metabolic-stress apoptosis, and — via the BARS branch — disturbing Golgi-dependent neuronal polarization and migration. The clinical core is the eponymous tetrad: **muscular hypotonia, ataxia, global developmental delay/intellectual disability, and tooth enamel defects**, with **cerebellar atrophy** on MRI. Fewer than ~20 patients have been reported worldwide. There is no disease-specific therapy; management is supportive.

> ⚠ **Gene disambiguation note:** This report concerns **MONDO:0060666 / OMIM #617915**, the ***CTBP1*-related** disorder HADDTS. It is distinct from *CTBP2*-related phenotypes; the systematic review found no confirmed neurodevelopmental disorder from heterozygous *CTBP2* variants ( 36331689 ).

## 1. Disease Information

**Overview.** HADDTS is a monogenic (Mendelian) pediatric neurodevelopmental syndrome. It was first delineated in 2016 when four unrelated patients with developmental delay, intellectual disability, failure to thrive, hypotonia, ataxia and tooth-enamel defects were each found to carry the same de novo *CTBP1* missense variant ( 27094857 ): *"we have identified four patients with similar phenotypes of developmental delay, intellectual disability, failure to thrive, hypotonia, ataxia, and tooth enamel defects who all have the same de novo R331W missense variant in C-terminal binding protein 1 (CTBP1)"* ( 27094857 ).

**Key identifiers.** - **MONDO:** MONDO:0060666 - **OMIM:** #617915 (Hypotonia, ataxia, developmental delay, and tooth enamel defect syndrome; HADDTS) - **Gene:** *CTBP1*, HGNC:2494, NCBI Gene 1487, Ensembl ENSG00000159692, UniProt Q13363; cytoband 4p16.3 - **Orphanet:** CTBP1-related neurodevelopmental disorder (ultra-rare; ORPHA designation under CTBP1-related conditions) - **ICD-10:** no specific code (mapped generically, e.g., Q87.8 / F79); **ICD-11:** no specific code (e.g., LD2F.1Y / 6A00.Z used generically) - **MeSH:** no dedicated descriptor; indexed under Neurodevelopmental Disorders / Intellectual Disability - **Data source type: Disease-level aggregation of individual published patient case reports/series** (not EHR registry data); evidence is predominantly human clinical case reports supplemented by in vitro and model-organism functional studies.

**Synonyms / alternative names:** HADDTS; CTBP1-related neurodevelopmental disorder; CTBP1-related intellectual disability; (historical) "R331W/R342W CTBP1 syndrome."

## 2. Etiology

**Primary cause — genetic.** HADDTS is caused by **heterozygous pathogenic variants in *CTBP1***, almost always arising **de novo** (P 42589715, P 27094857). There is no environmental, infectious, or multifactorial etiology; it is a classic de novo dominant Mendelian disorder.

**Genetic risk factors.** - **Causal variant (recurrent hotspot):** *CTBP1* **p.Arg342Trp** (c.991C>T or c.1024C>T depending on transcript; = legacy p.R331W). 13/14 individuals in the systematic review carried this variant (P 36331689). - **Second reported variant: c.1315_1316delCA, p.Gln439ValfsTer84** (frameshift) in one individual (P 36331689). - **Expanded allelic spectrum:** a novel **c.107G>C, p.Arg36Pro** variant, also in the PLDLS cleft, produced a *distinct* neurodevelopmental disorder not fully meeting HADDTS criteria (P 41674141). - **Modifier genes:** none established. *CTBP2* is a paralog with overlapping function and is a plausible genetic modifier (redundancy), but no human modifier data exist. - **Primary risk factor for occurrence:** advanced parental age is the generic correlate of de novo mutation burden (not specifically demonstrated for *CTBP1*).

**Environmental risk factors / lifestyle / infectious agents: None known / not applicable.** No toxin, exposure, diet, or pathogen is implicated. (CTD: no curated environmental interaction causing HADDTS.)

**Protective factors (genetic or environmental): None identified / not applicable.**

**Gene–environment interactions.** Not established clinically. Mechanistically, CtBP1 is a **metabolic/redox sensor** (NAD(H)-dependent), and patient fibroblasts show heightened sensitivity to **glucose deprivation** ( P 31041561 ) — a cell-level gene×metabolic-state interaction, but no organism-level GxE is documented.

## 3. Phenotypes

Evidence base: human clinical case reports/series ( P 27094857 P 31041561 P 36331689 ). P 42704868 Frequencies are approximate given the very small cohort (<20 patients).

**Core tetrad (present in most/all; "consistent CtBP1-related phenotypes of intellectual disability, ataxia, hypotonia, and tooth enamel defects present in most patients," ):** P 31041561

| Phenotype | Type | Onset | Severity | Course | Freq. | HPO |
|---|---|---|---|---|---|---|
| Intellectual disability / global developmental delay | cognitive/ behavioral | neonatal–infancy | moderate–severe | stable/non-progressive | ~100% | HP:0001249; HP:0001263 |
| Muscular hypotonia ("floppy infant") | clinical sign (neuromuscular) | neonatal | moderate–severe | often improves to spasticity/ stable | ~100% | HP:0001252; HP:0001319 |
| Ataxia (gait/ truncal) | clinical sign (cerebellar) | childhood | moderate | stable | high (most) | HP:0001251 |
| Tooth enamel defect (enamel hypoplasia/ amelogenesis imperfecta-like) | physical/dental | childhood (as teeth erupt) | variable | stable | high (defining) | HP:0000682; HP:0006297 |

**Additional/variable features reported:** - Cerebellar atrophy / volume loss on MRI (HP:0001272; cerebellar vermis atrophy HP:0006855) — *"cerebellar volume loss"* ( P 36331689 ); *"cerebellar atrophy on brain imaging"* ( P 42704868 ). - Delayed/absent speech, speech disorder (HP:0000750; HP:0001344). - Failure to thrive / feeding difficulties (HP:0001508; HP:0011968) ( P 27094857 ). - Facial dysmorphism (HP:0001999) ( P 42704868 ). - Constipation (HP:0002019) ( P 42704868 ).

- Congenital heart disease — atrial septal defect (HP:0001631) reported in the p.R36P case ( P 41674141 ).
- Behavioral features of severe NDD (variable).

**Quality-of-life impact.** High burden: nonverbal or limited speech, dependence for daily activities, mobility limited by hypotonia/ataxia, and dental morbidity (pain, caries susceptibility from defective enamel). No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disorder; QoL inferred from severe-NDD phenotype.

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## 4. Genetic / Molecular Information

**Causal gene.** ***CTBP1*** (HGNC:2494; OMIM gene *602618*; disease OMIM #617915). Encodes a 440-aa NAD(H)-dependent d-isomer–specific 2-hydroxyacid dehydrogenase-family transcriptional corepressor (UniProt Q13363).

**Pathogenic variants.** - **p.Arg342Trp (recurrent hotspot):** c.991C>T / c.1024C>T; **missense**; **Pathogenic (ACMG)**; recurrent de novo; in the PLDLS-binding cleft. ~90% of cases ( P 36331689 P 27094857 P 31041561 ). P 42704868
- **p.Gln439ValfsTer84:** c.1315_1316delCA; **frameshift**; **Pathogenic**; single case ( P 36331689 ).
- **p.Arg36Pro:** c.107G>C; **missense**; PLDLS cleft; distinct/atypical NDD ( P 41674141 ).
- **Allele frequency:** all are **absent from population databases** (gnomAD) — consistent with de novo pathogenic variants in an essential, highly constrained gene (*CTBP1* is loss-of-function intolerant). **Not found as common polymorphisms.** - **Somatic vs germline: germline** (constitutional de novo); one case of **maternal germline mosaicism** ( P 36331689 ).

(Note: somatic *CTBP1* dysregulation/overexpression is separately implicated in cancer — not relevant to HADDTS germline disease.) - **Functional consequence: Dominant-negative / interfering gain-of-abnormal-function** at the PLDLS cleft rather than simple loss of function. Supported by (i) viability of *Ctbp1*-null mice ( P 12101226 ); (ii) reduced partner binding by the mutant plus transcriptional deregulation ( P 31041561 ). The variant retains dimerization/NAD-binding but loses high-affinity partner recruitment, enabling it to poison shared CtBP1/CtBP2 complexes.

**Modifier genes.** None proven; *CTBP2* redundancy is the leading candidate buffer ( P 12101226 ). P 12508276

**Epigenetic information.** CtBP1 is itself an **epigenetic effector** — it scaffolds HDAC1/HDAC2 and LSD1/KDM1A to deacetylate and demethylate histones at target enhancers ( P 41105133 ). No disease-specific DNA-methylation "episignature" has yet been published for HADDTS (a plausible future diagnostic avenue, not applicable currently).

**Chromosomal abnormalities.** None; HADDTS is a single-nucleotide/small-indel disorder. Chromosomal microarray is typically normal.

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## 5. Environmental Information

**Environmental factors, lifestyle factors, infectious agents: None / not applicable.** HADDTS is a purely genetic de novo dominant disorder with no established environmental contribution. The only environment-adjacent biology is cellular **metabolic/redox state** (glucose/NAD(H)), which modulates CtBP1 activity and patient-cell apoptosis sensitivity

( P 31041561 , P 38626254 ) — relevant mechanistically, not as a population exposure.

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## 6. Mechanism / Pathophysiology

### Ordered causal chain (initiating lesion → clinical manifestation)

1. A **de novo heterozygous *CTBP1* variant** (predominantly p.Arg342Trp) arises in the germline → an altered CtBP1 protein with a defective **PLDLS-binding cleft**.
2. The mutant **leads to** reduced recruitment of PLDLS-motif corepressor partners and chromatin-modifying enzymes (HDAC1/2, LSD1) to CtBP1 (*demonstrated*: reduced interaction with chromatin-modifying factors, P 31041561 ; complex identity, P 41105133 ).
3. Because CtBP1 dimerizes with/shares partners with CtBP2, the mutant **acts as a dominant-negative**, poisoning the combined CtBP1/CtBP2–HDAC–LSD1 repressor activity (*inferred* from *Ctbp1*-null mouse viability + dosage sensitivity, P 12101226 / 12508276).
4. This **results in** genome-wide **transcriptional deregulation** of genes controlling multiple cellular processes (*demonstrated* by transcriptomics, P 31041561 ).

**Branch A (nuclear / chromatin):** Deregulated neurogenic enhancer programs (PRDM16–CtBP–HDAC1/2–LSD1) **lead to** impaired neural progenitor proliferation and **delayed migration/positioning of cortical neurons** (P 41105133 ). P 32169478

**Branch B (cytoplasmic / BARS):** Impaired CtBP1-S/**BARS**-mediated **Golgi membrane fission leads to** defective post-Golgi trafficking, disturbing neuronal polarization, axon/dendrite elongation and the multipolar-to-bipolar transition (*BARS requirement demonstrated by RNAi,* P 35455998 ; dual function, P 41674141 ; whether HADDTS alleles impair BARS fission specifically is *inferred*).

**Branch C (metabolic / apoptotic):** As an NAD(H) sensor, mutant CtBP1 **results in** heightened sensitivity to metabolic stress — glucose deprivation strongly activates the BH3-only gene **NOXA/PMAIP1**, enhancing apoptosis (P 31041561 ); a functional case links the disorder to **impaired mitochondrial bioenergetics** (P 42123581 ).

5. Branches A–C **converge** to **reduce dendritic number/length, disrupt dendritic arborization and spine formation, and lower excitatory synaptic transmission** (*demonstrated in vitro and in vivo,* P 41674141 ).

6. These cellular defects **result in** maldevelopment of **cerebral cortex and cerebellum → cerebellar atrophy**, and clinically **hypotonia, ataxia, developmental delay/intellectual disability** (P 36331689 ). P 42704868

7. In parallel, CtBP1 corepressor dysfunction in **enamel-forming epithelium (ameloblasts) leads to** the **tooth enamel defect** (*mechanism inferred*; ameloblast-specific data not yet published).

## Detail by category

- **Molecular pathways:** CtBP corepressor axis (PLDLS-partner → HDAC1/2 + LSD1/KDM1A, recruited via PRDM16, ZEB1/δEF1, CTCF) ( P 10567582 ); P 41105133 NAD(H)/redox-glycolysis signaling controlling CtBP oligomerization ( P 38626254 ); P 41972124 intersections with Wnt-regulated repression (CtBP is a Wnt-pathway corepressor). GO: negative regulation of transcription by RNA Pol II (GO:0000122); transcription corepressor activity (GO:0003714).
- **Cellular processes:** neural progenitor proliferation & migration (GO:0001764), dendrite development (GO:0016358), dendritic spine morphogenesis (GO:0060997), synaptic transmission (GO:0007268), apoptosis (GO:0006915), Golgi vesicle budding/fission (GO:0048194).
- **Protein dysfunction:** loss of high-affinity PLDLS partner binding with retained scaffold → **dominant-negative**; two functional modules = NAD(H)-binding Rossmann fold (intact) + PLDLS cleft (disrupted) ( P 38711119 ).
- **Metabolic changes:** NAD(H)/glucose sensitivity; impaired mitochondrial bioenergetics ( P 42123581 ). P 31041561
- **Immune involvement:** not a feature of HADDTS (CtBP1's inflammasome-transrepression role is a separate vascular context, P 38626254 ).

- **Tissue damage mechanism:** developmental (maldevelopment) rather than degenerative, though "cerebellar atrophy" implies some neuronal loss/under-growth.
- **Molecular profiling available:** transcriptomics in glioblastoma lines expressing wt vs W342 CtBP1 ( P 31041561 ); unbiased proteomics of mutant interactome ( P 31041561 ); isogenic iPSC-derived neuronal models ( P 41459558 ); no HADDTS metabolomics/lipidomics/spatial datasets yet.
- **Cell types (CL):** glutamatergic/excitatory cortical neuron (CL:0000679), pyramidal neuron, neural progenitor cell (CL:0011020), cerebellar neurons (Purkinje/granule), ameloblast (CL:0000059, enamel).
- **Subcellular compartments (GO CC):** nucleus (GO:0005634), Golgi apparatus (GO:0005794), mitochondrion (GO:0005739).

## 7. Anatomical Structures Affected

- **Organ/system level — primary: central nervous system** (UBERON:0001017) — **cerebral cortex** (UBERON:0000956) and **cerebellum** (UBERON:0002037, with cerebellar atrophy/ vermis involvement); **teeth/dental enamel** (UBERON:0001752 tooth; UBERON:0001751 enamel). Body systems: **nervous**, **musculoskeletal/neuromuscular** (hypotonia), **stomatognathic/dental**.
- **Secondary involvement: digestive** (feeding difficulty, constipation), **cardiovascular** (atrial septal defect in one case), growth (failure to thrive).
- **Tissue/cell level: nervous tissue** (cortical & cerebellar neurons, neural progenitors); **dental epithelium** (ameloblasts → enamel matrix); skeletal **muscle** secondarily (central hypotonia, not primary myopathy).
- **Subcellular:** nucleus (chromatin regulation), Golgi apparatus (BARS fission), mitochondria (bioenergetics).

- **Localization / lateralization: bilateral, symmetric** CNS involvement; cerebellar atrophy is typically diffuse/midline (vermian).

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## 8. Temporal Development

- **Onset: congenital/neonatal** — hypotonia and feeding problems present from birth ("floppy baby,"
  ); P 42704868
  developmental delay evident in infancy. Pattern: **chronic, insidious** (developmental, not acute).
- **Progression: largely static/non-progressive neurodevelopmental** course; disability is lifelong. Enamel defects manifest as dentition erupts. Cerebellar atrophy documented on imaging; whether it is static or slowly progressive across the small cohort is **not firmly established** (natural-history data limited).
- **Disease course pattern:** stable/chronic lifelong; not relapsing-remitting or episodic.
- **Critical period / window of opportunity: corticogenesis (prenatal) and early postnatal synaptic maturation** — the mechanistic steps (neuron migration, dendrite/spine formation) occur developmentally, so the therapeutic window is early; postnatal reversal is likely limited.
- **Remission:** none (no spontaneous or treatment-induced remission reported).

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## 9. Inheritance and Population

- **Epidemiology: ultra-rare**; **<20 patients reported worldwide** as of 2026 — *"This is the 18th reported case of this ultra-rare disorder"*
  ( P 42704868 );
  *"only 20 published cases"*
  ( P 42123581 ).
  **Prevalence/incidence not quantifiable** (no population estimate; effectively <1/1,000,000).

- **Inheritance pattern: autosomal dominant** (OMIM #617915), essentially all cases **de novo** (P 42589715). P 27094857.
- **Penetrance:** presumed **complete** in heterozygous carriers of the pathogenic variant (all reported carriers affected).
- **Expressivity: variable** (core tetrad consistent; additional features e.g., cardiac, dysmorphism vary; distinct allele p.R36P gives an atypical presentation).
- **Genetic anticipation:** not applicable (not a repeat-expansion disorder).
- **Germline mosaicism: documented** (one maternal germline mosaicism case, P 36331689) → **non-zero recurrence risk** for apparently unaffected parents.
- **Founder effects / consanguinity / carrier frequency: not applicable** (recurrent de novo dominant; the recurrence reflects a mutational **hotspot**, not a founder haplotype; consanguinity irrelevant; no carrier screening relevance).
- **Population demographics:** reported across diverse populations (US, Iran [PMID 34608870], Poland [PMID 42704868], others) → **no ethnic predilection**; recurrence is hotspot-driven, not geographic. **Sex ratio:** apparent male excess (10/14 males, P 36331689) but likely ascertainment artifact given tiny N. **Age distribution:** children and young adults (diagnosed in childhood; oldest reported ~20 years).

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## 10. Diagnostics

- **Primary diagnostic modality — genomic sequencing.** Diagnosis is molecular: **trio whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** identifies the de novo *CTBP1* variant — *"Exome sequencing is an effective way to identify genetic causes of etiologically heterogeneous conditions such as developmental delay and intellectual disabilities"* (P 27094857); subsequent cases diagnosed by WES

( P 34608870 ) and genotype-to-phenotype large-scale testing ( P 42704868 ).

**Single-gene/targeted *CTBP1* testing** or **NDD gene panels** including *CTBP1* are appropriate when clinically suspected. Confirmation by Sanger; parental testing to establish de novo status.

- **Low-yield / normal tests: chromosomal microarray (CMA)** and **karyotype** are typically normal (no structural lesion); FISH, mtDNA testing, and repeat-expansion testing are **not indicated**.
- **Supportive clinical tests:**
- **Brain MRI:** cerebellar atrophy/volume loss (HP:0001272) ( P 36331689 ). P 42704868
- **Dental examination/radiographs:** enamel hypoplasia/defects.
- **Echocardiography:** to evaluate congenital heart defect when present.
- **Metabolic workup:** generally non-diagnostic but excludes mimics; patient fibroblasts show glucose-deprivation apoptosis sensitivity (research finding, P 31041561 ), and one report notes mitochondrial bioenergetic impairment ( P 42123581 ) — not yet a clinical biomarker.
- **Biomarkers:** no validated circulating/biochemical biomarker; the **genetic variant itself is the diagnostic marker**. A DNA-methylation episignature is a plausible future tool (not available).
- **Clinical/diagnostic criteria:** no formal consensus criteria; diagnosis = compatible phenotype (hypotonia + ataxia + DD/ID + enamel defect ± cerebellar atrophy) **plus a pathogenic *CTBP1* variant**. Note: the atypical p.R36P patient did **not** fulfill HADDTS criteria despite a pathogenic *CTBP1* variant ( P 41674141 ), underscoring phenotype–genotype nuance.

- **Differential diagnosis:** other causes of the hypotonia–ataxia–ID–enamel constellation, e.g., **Kohlschütter–Tönz syndrome** (*ROGDI*; epilepsy + amelogenesis imperfecta), other **amelogenesis imperfecta** syndromes, **congenital disorders of glycosylation**, **cerebellar hypoplasia/atrophy syndromes**, and other monogenic NDDs — distinguished by gene-level testing.
- **Screening (asymptomatic):** not applicable (de novo; no carrier/newborn screening); **cascade testing** only relevant for recurrence counseling given mosaicism.

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## 11. Outcome / Prognosis

- **Survival/mortality: No disease-specific mortality data**; HADDTS is **not reported as life-limiting per se**. Life expectancy is presumably determined by severity of disability and complications (feeding, aspiration, cardiac anomaly when present) rather than by a defined lethal course. Oldest reported patient ~20 years ( P 41674141 ).
- **Morbidity/disability: high, lifelong** — intellectual disability (often moderate–severe), motor disability from hypotonia and ataxia, speech impairment, and dental morbidity. Most patients require lifelong caregiving support.
- **Disease course:** chronic, static neurodevelopmental disability; complications include feeding difficulty/failure to thrive, constipation, dental caries/pain from enamel defects, and cardiac issues in a minority.
- **Recovery potential:** developmental gains with early intervention/therapy are possible, but **the underlying neurodevelopmental deficit is not reversible**.
- **Prognostic factors:** genotype (classic p.R342W vs atypical alleles), severity of cerebellar atrophy, presence of cardiac/other systemic anomalies, and feeding status. **No validated prognostic biomarkers.** QoL/ICF-coded disability instruments not formally applied.

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## 12. Treatment

**There is no disease-specific or disease-modifying therapy for HADDTS. Management is entirely supportive and multidisciplinary** (evidence: absence of any reported pharmacologic/gene/cell/RNA therapy or HADDTS-specific trial; standard severe-NDD care).

- **Pharmacotherapy: none specific.** Symptomatic agents as needed (e.g., for constipation; anti-epileptics only if seizures occur). **Pharmacogenomics:** not applicable.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno): none approved or in trials.** Note a conceptual mismatch: existing **CtBP dehydrogenase inhibitors** (MTOB, JW-98, Comp.11) were developed to *reduce* CtBP oncogenic activity in cancer (P 41353240 ) (P 38711119 ) — the **opposite** of what a loss-of-corepression developmental disorder would require; they are **not** HADDTS therapeutics. Because the mechanism is dominant-negative, **allele-selective knockdown (ASO/siRNA against the mutant allele)** is a theoretically rational but **unproven/experimental** future direction (NCIT: Antisense Oligonucleotide Therapy C1516; Gene Therapy C15262).
- **Surgical/interventional:** as indicated for complications (e.g., congenital heart defect repair; dental restorative procedures/crowns for enamel defects).
- **Supportive & rehabilitative (mainstay): physical therapy** (NCIT:C15917) for hypotonia/ataxia/mobility; **occupational therapy** (NCIT:C15224); **speech-language therapy** (NCIT:C15348) and AAC; **nutritional support/feeding management** (NCIT:C15491) for failure to thrive; **restorative/preventive dental care** for enamel defects; management of constipation; early-intervention/special-education programs. NCIT: Supportive Care (C15277).
- **Treatment outcomes / adverse events:** supportive measures improve function/comfort; no curative outcome. **Treatment strategy:** individualized, symptom-directed multidisciplinary care; genotype-guided therapy not yet available (**personalized medicine** = **experimental allele-specific approaches only**).

## 13. Prevention

- **Primary prevention: none possible** for the de novo event itself (no modifiable risk factor).
- **Secondary prevention:** early developmental surveillance and early-intervention therapy to optimize function; early dental and cardiac evaluation to pre-empt complications (tertiary prevention of complications).
- **Genetic screening / reproductive options:** once a familial/proband variant is identified, **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** are available for future pregnancies; recurrence risk is low but **non-zero due to documented germline mosaicism** ( P 36331689 ).
- **Counseling: genetic counseling** (NCIT:C15587) is central — explaining de novo dominant inheritance, low-but-nonzero recurrence risk (mosaicism), and reproductive options.
- **Immunization / behavioral / public-health / environmental / prophylaxis: not applicable** (no infectious or environmental component).

## 14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *CTBP1* is deeply conserved. Orthologs: **mouse *Ctbp1*** (NCBI Gene 13016; *Mus musculus*, NCBI:txid10090), **rat *Ctbp1*** (*Rattus norvegicus*, NCBI:txid10116), zebrafish *ctbp1* (*Danio rerio*, NCBI:txid7955), *Drosophila* **CtBP** (NCBI:txid7227), and a plant homolog **Angustifolia** ( P 12508276 ). The paralog **CTBP2/*Ctbp2*** (NCBI Gene 13017 mouse) is partially redundant.
- **Natural disease in other species / breeds / VBO / zoonosis: None reported.** No naturally occurring HADDTS-equivalent disease, no breed predisposition (VBO: not applicable), and **no zoonotic/transmission** relevance — it is a non-communicable germline disorder.
- **Comparative biology / conservation:** disease mechanisms are evolutionarily conserved — CtBP corepressor function and PLDLS-partner recruitment are conserved from flies to humans ( P 10567582 )

), P 12508276 and CtBP controls neurodevelopment across vertebrates, enabling model-organism study.

## 15. Model Organisms

- **Mouse (*Mus musculus, NCBI:txid10090) — knockouts:** Ctbp1-*null mice are small but viable and fertile, whereas* **Ctbp2*-null mice are embryonic lethal (~E10.5)** with axial-patterning, heart-morphogenesis and neural-development defects; compound alleles show **dosage-sensitive** phenotypes (P 12101226 ). P 12508276 **Interpretation:** *Ctbp1* loss alone is well tolerated → argues the human p.R342W is **dominant-negative**, not haploinsufficient. (Available via MGI/IMSR.)
- **Mouse — in vivo corticogenesis models: in utero electroporation** of CtBP2 siRNA perturbs neural stem/progenitor proliferation and migration ( P 32169478 ); the **PRDM16–CtBP1/2–HDAC1/2–LSD1** complex guides upper-layer neuron migration ( P 41105133 ); acute in vivo expression of human **p.R342W and p.R36P** causes excitatory-neuron migration delay and disrupted dendritic arborization/spine formation ( P 41674141 ). **Strong recapitulation** of the human neuromigration/dendritic phenotype.
- **Primary neurons / in vitro:** hippocampal neuron cultures show reduced dendrite number/length with mutant CtBP1 ( P 41674141 ) and impaired axon/dendrite growth with BARS knockdown ( P 35455998 ).

- **Patient-derived & iPSC models:** patient **dermal fibroblasts** (glucose-deprivation apoptosis, NOXA induction; P 31041561); **isogenic iPSC-derived neurons** modeling the heterozygous mutation (P 41459558); **glioblastoma lines** expressing wt vs W342 CtBP1 for transcriptomics (P 31041561).
- **Model limitations:** no single model reproduces the **full human tetrad** (especially the **tooth enamel defect**, which lacks a dedicated published model); mouse knockouts model *loss* rather than the *dominant-negative* allele; cerebellar/ataxia phenotype not fully characterized in a germline knock-in mouse (a **p.R342W knock-in mouse is a key gap**).
- **Resources:** MGI (mouse), IMSR/IMPC (alleles), ZFIN (zebrafish), FlyBase (Drosophila), Cellosaurus (iPSC/cell lines).

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## Supported vs Refuted Hypotheses

**Supported:** 1. HADDTS is caused by de novo heterozygous *CTBP1* variants, dominated by the recurrent PLDLS-cleft hotspot p.Arg342Trp (P 27094857; P 36331689).
2. The variant acts via a **dominant-negative** mechanism (Ctbp1-null mice viable; mutant loses partner binding but retains scaffold) (P 12101226; P 31041561).
3. Pathogenesis converges on **corticogenesis**: impaired neuron migration, dendrite/spine maturation, and excitatory transmission (P 41674141

P 41105133 ). P 32169478

4. CtBP1 couples **metabolic/redox state** to transcription, explaining glucose-stress apoptosis ( P 31041561 P 38626254 ). P 41972124

5. A **second (BARS/Golgi-fission) branch** contributes to neuronal maldevelopment ( P 35455998 ). P 41674141

**Refuted / not supported:** - Simple **haploinsufficiency** as the mechanism (refuted by viable *Ctbp1*-null mice). - A confirmed ***CTBP2***-related equivalent neurodevelopmental disorder (not identified; ). P 36331689

- Any environmental/infectious etiology or population risk factor (none exist).

## Limitations and Future Directions

- **Tiny evidence base** (<20 patients) → frequencies, sex ratio, prognosis and natural history are imprecise.
- **Enamel-defect mechanism is inferred**, not experimentally demonstrated (no ameloblast model).
- **No germline p.R342W knock-in animal** to model the dominant-negative allele and the full tetrad (key gap).
- **No disease-specific therapy**; allele-selective silencing is a rational but untested avenue given the dominant-negative mechanism.
- **No HADDTS-specific omics atlas** (metabolomics, single-cell/spatial of patient brain) and **no methylation episignature** yet.

- Genotype–phenotype expansion (e.g., p.R36P atypical disorder) suggests the *CTBP1* phenotypic spectrum is broader than the classic tetrad and warrants systematic re-delineation.

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## Key references (PMID)

27094857 (original 4-patient delineation) · 31041561 (7-patient cohort + mechanism/NOXA) · 36331689 (systematic review, 14 cases) · 42589715 (variant spectrum review) · 41674141 (R36P; in vivo corticogenesis; dual function) · 42704868 (18th case; floppy baby; dominant-negative) · 42123581 (mitochondrial bioenergetics) · 41459558 (isogenic iPSC model) · 34608870 (Iranian WES case) · 12101226 & 12508276 (Ctbp1/2 mouse genetics) · 32169478 & 41105133 (CtBP in cortical neurogenesis/migration) · 35455998 (BARS neuronal trafficking) · 10567582 (PLDLS corepressor mechanism) · 38626254, 41972124, 38711119, 41353240 (NAD(H)-sensing, structure, inhibitors).

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